

Cooperative Game Theory

Definition

Cooperative Game Theory is a branch of game theory that studies situations in which **players can communicate, negotiate, and form binding agreements (coalitions)** to achieve better outcomes than they could individually.

Unlike **non-cooperative game theory**, where each player acts independently, cooperative game theory focuses on **teamwork, collaboration, and the fair distribution of the gains generated by cooperation.**

Formal Definition

A **cooperative game** is represented by:

$$(N, v)$$

Where:

- **N** = Set of players.
- **v(S)** = Value (worth) that any coalition $S \subseteq N$ can achieve together.

The function **v** is called the **characteristic function**, assigning a value to every possible coalition.

Key Concepts

1. Coalition

A **coalition** is a group of players who cooperate to improve their collective outcome.

2. Characteristic Function

The characteristic function tells us the total payoff a coalition can earn.

3. Transferable Utility (TU)

Players can freely share or redistribute profits.

4. Non-Transferable Utility (NTU)

Utility cannot be freely transferred.

Core (Solution Concept)

The **Core** is the set of payoff distributions where **no subgroup of players has an incentive to leave the grand coalition**, because they cannot do better on their own.

A payoff allocation is in the Core if:

- Everyone receives at least what they could earn alone.
- Every coalition receives at least its worth.
- The total allocated equals the value of the grand coalition.

If an allocation is in the Core, cooperation is considered **stable**

Shapley Value

The **Shapley Value** provides a fair way to distribute the total payoff based on each player's average contribution across all possible coalitions.

Formula

$$\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|! (n - |S| - 1)!}{n!} [v(S \cup \{i\}) - v(S)]$$

Intuition

Each player is rewarded according to their **marginal contribution** to every coalition they join.

Example

Three friends start a business.

Annual profit:

₹120 lakh

Individual contributions:

- A: Marketing
- B: Technology
- C: Finance

The Shapley Value computes a fair profit share by considering each person's average added value across all possible team formations.

Nash Bargaining Solution

When two players negotiate, the Nash Bargaining Solution identifies an agreement that is:

- Efficient
- Fair
- Better than disagreement for both players

It maximizes:

$$(U_1 - d_1)(U_2 - d_2)$$

where:

- U_1, U_2 are the agreed utilities.
- d_1, d_2 are the disagreement (fallback) utilities.

Types of Cooperative Games.

Type	Description
Transferable Utility (TU)	Utility can be redistributed.
Non-Transferable Utility (NTU)	Utility cannot be transferred.
Superadditive Game	Cooperation always increases total payoff.
Convex Game	Incentives to cooperate grow as coalitions become larger.
Simple Game	Coalitions either win or lose (e.g., voting).

Examples

Real-Life Examples

1. Business Alliance
2. Political Coalition
3. Supply Chain Collaboration
4. Joint Research Projects

Advantages

- Encourages collaboration and trust.
- Produces outcomes that can improve overall welfare.
- Offers fair methods for allocating joint gains (e.g., Shapley Value).
- Supports decision-making in economics, politics, business, and international relations.
- Helps analyze coalition formation and stability.

Limitations

- Assumes players can make and enforce binding agreements.
- Fair allocation can become computationally difficult with many players.
- Coalition negotiations may involve strategic behavior not fully captured by the model.
- Some solution concepts (like the Core) may be empty, meaning no stable allocation exists.

<u>Feature</u>	<u>Cooperative Game Theory</u>	<u>Non-Cooperative Game Theory</u>
Agreements	Binding agreements allowed	No binding agreements
Cooperation	Players form coalitions	Players act independently
Main Goal	Maximize collective payoff and divide it fairly	Maximize individual payoff
Key Concepts	Coalitions, Core, Shapley Value, Nash Bargaining	Nash Equilibrium, Dominant Strategy
Examples	Business alliances, coalition governments, joint ventures	Price competition, auctions, prisoner's dilemma